

Question Paper

Exam Date & Time: 06-Jun-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

SECOND SEMESTER B.TECH. DEGREE EXAMINATIONS - JUNE 2025
SUBJECT: ELE 1071/ELE_1071 - BASIC ELECTRICAL TECHNOLOGY
(MAKE-UP)

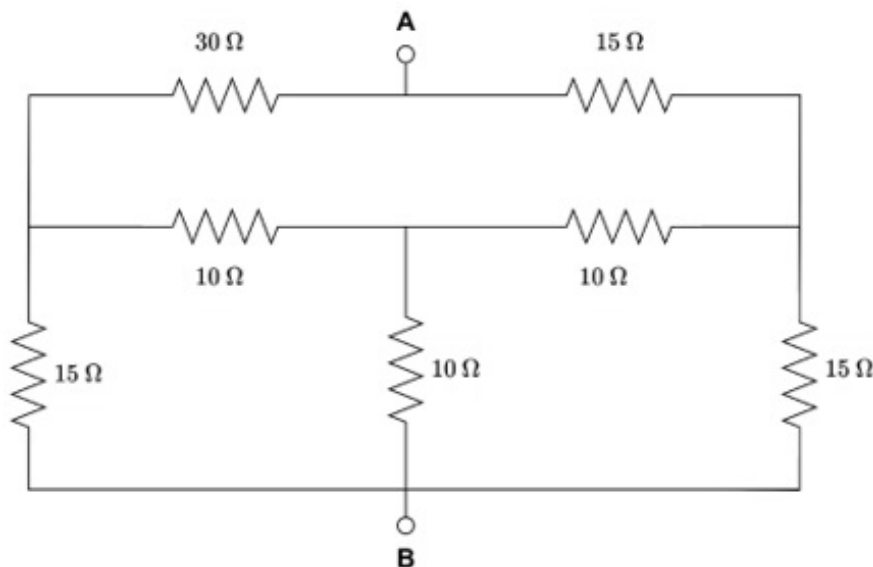
Marks: 50

Duration: 180 mins.

Answer all the questions.

1A) Determine the equivalent resistance between terminals A and B.

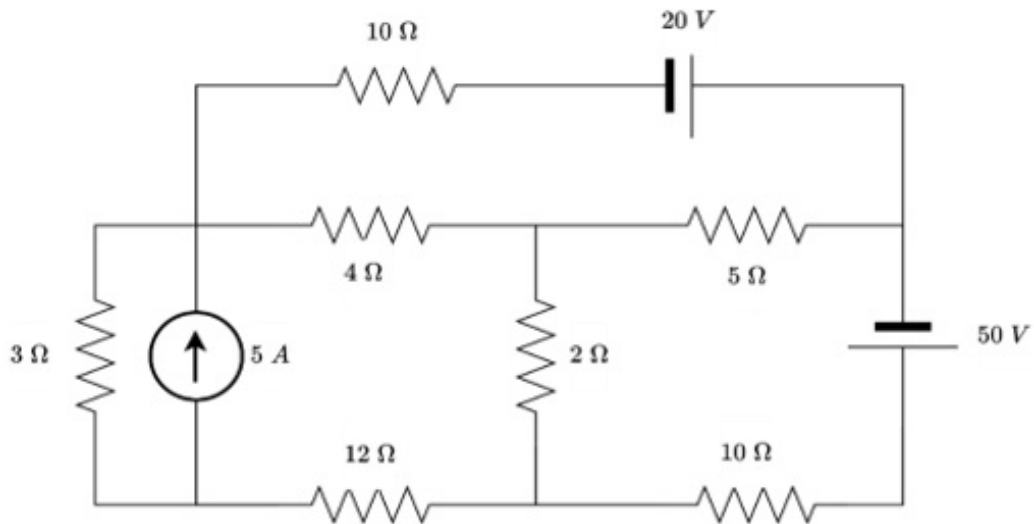
(3)



Course Outcome: "CO1" / Cognitive Level: "Application"

1B) Using Superposition theorem, determine the current through 12 Ω resistance.

(7)



Course Outcome: "CO1" / Cognitive Level: "Application"

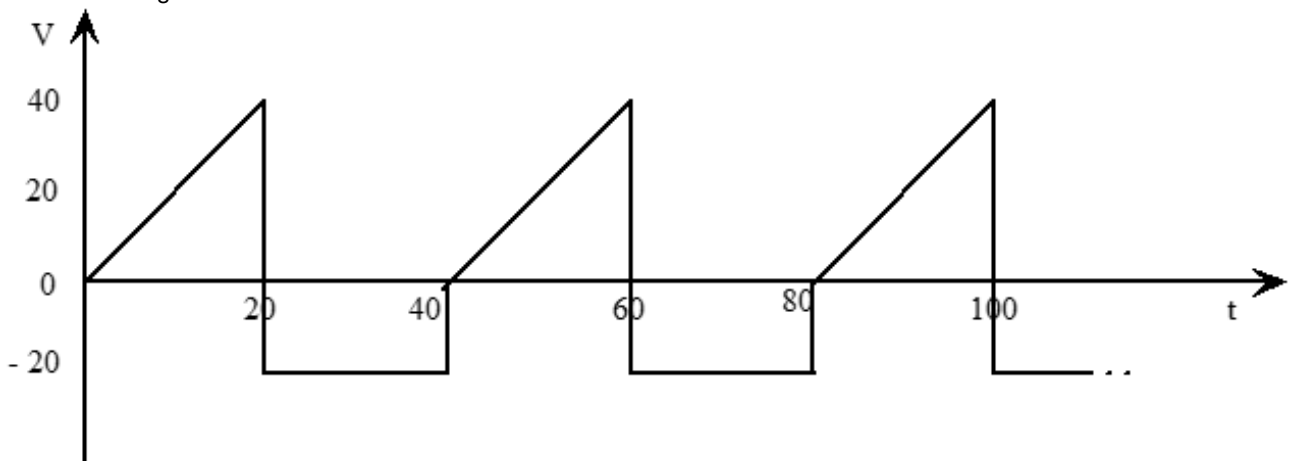
- 2A) A single-phase ac circuit with applied voltage $v(t) = 200 \sin 377 t$ volts, draws a current of $i(t) = 8 \sin (377t - 30^\circ)$ amperes. Find (i) impedance of the circuit (ii) Circuit parameters (iii) Power and power factor associated. (5)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 2B) A 100Ω resistor and a 0.5 H inductor are connected in series with a 230 V , 50 Hz AC supply. If a capacitor has to be added in parallel to the supply, calculate the value of the capacitor to enhance the overall power factor to 0.9 lagging. (5)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 3A) Find the average and rms value of the waveform shown below. (4)



Course Outcome: "CO2" / Cognitive Level: "Application"

- 3B) An unbalanced three phase star connected load, is supplied with a three phase four wire 415 V , 50 Hz , RYB supply. If $Z_R = 6 + j8 \Omega$, $Z_Y = 12 - j5 \Omega$ and $Z_B = 20 \Omega$, calculate the neutral current. Take V_{RN} as reference. (6)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 4A) Two-watt meters are connected to measure three phase power of a balanced star connected load, drawing a phase current of 20 A , at 0.7 power factor lagging. Determine the reading of the watt meters if the supply voltage is 400 V . (4)

- 4B) A ring of 30 cm mean diameter with an airgap of 1 mm is made using a cylindrical mild steel rod of diameter 2.5 cm. When it is magnetized with a coil carrying a current of 6 A, the flux produced in the core is 3 mWb. Taking relative permeabilities of mild steel as 2000, calculate (6)
- Flux density within the core
 - Total reluctance of the ring
 - Number of turns in the coil
 - Magnetomotive force

- 5A) The combined inductance of two coils connected in series is either 1.2 H or 0.2 H, depending on the relative directions of the current in the coils. If self-inductance of one of the coils is 0.4 H, determine (3)
- Self-inductance of the other coil.
 - Mutual inductance.
 - Coupling coefficient.

- 5B) Clearly indicating the voltage levels stage, explain briefly the role of a transformer in an Electric Power System. (4)

- 5C) With a neat block diagram, explain the working of a digital energy meter. (3)

-----End-----

Event Name			Subject
MIT MPL and BLR- BTech-MSc-MCA - II-IV and VI Sem - Makeup Examination - Jun 2025			ELE 1071-CHM:BASIC ELECTRICAL TECHNOLOGY
#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	14
3	3	CO 3	10
4	4	CO 4	9
5	5	CO 5	7
Total			50

* **NA** - CO Not Selected

Event Name			Subject
MIT MPL and BLR- BTech-MSc-MCA - II-IV and VI Sem - Makeup Examination - Jun 2025			ELE 1071-CHM:BASIC ELECTRICAL TECHNOLOGY
#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	7(14%)
3	Application	3	14(28%)
4	Analysis	6	29(58%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		11	50 (100%)